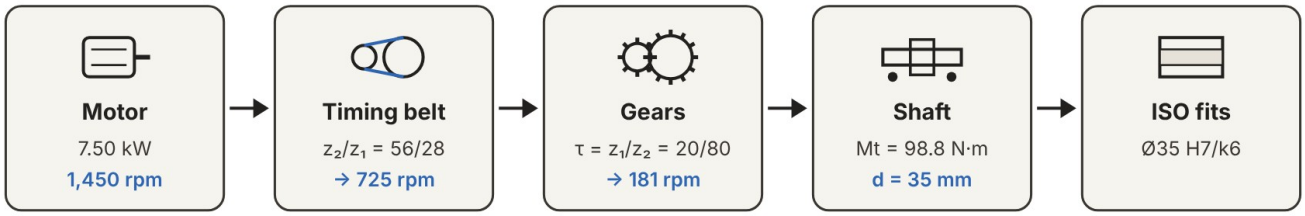
Complete transmission — preliminary sizing

Timing belt · Gears · Shaft · ISO fits

Contents

1. Transmission layout and power flow
2. Synchronous belt drive — preliminary sizing
3. Pair of cylindrical gears — preliminary sizing
4. Transmission shaft — preliminary sizing
5. Shaft-hole fit ISO 286

1 Transmission layout and power flow



Stage	Ratio	n in [rpm]	n out [rpm]	P [kW]	M in [N·m]	M out [N·m]
Timing belt	z_2/z_1 = 56/28	1,450	725	7.50	49.4	98.8
Gears	τ = z_1/z_2 = 20/80	725	181	7.50	98.8	395.1
Shaft	—	725	725	7.50	98.8	98.8

Consistency between modules

✓	Speed: belt output 725 rpm, pinion 725 rpm	consistent
✓	Power: belt 7.50 kW, gears 7.50 kW	consistent
✓	The shaft turns as the pinion shaft (725 rpm)	consistent
⚠	Mounting of the pinion: root Ø 39.4 mm on shaft Ø35 → rim under the teeth 2.2 mm < 2.5·m = 5.6 mm → make the pinion integral with the shaft (pinion shaft)	to be checked
✓	Power on the shaft 7.50 kW, upstream stage 7.50 kW	consistent
✓	Fit Ø 35 mm, shaft Ø 35 / 42 mm	consistent

Each module uses its own data: if a stage is not consistent, correct the values in the module and generate the report again.

Summary of the results

Chapter	Main result
2. Timing belt	HTD 8M (8 mm) · z_1 = 28, z_2 = 56 · C = 450.6 mm
3. Gears	z_1 = 20, z_2 = 80, m = 2.25 mm, L = 32.8 mm
4. Shaft	d = 35 mm
5. ISO fits	Ø35.00 H7/k6 · transition

Efficiencies neglected (constant power along the chain); speeds follow the transmission ratios: belt n2=n1z1/z2, gears n2=n1τ with τ=z1/z2; torque M=P/ω.

2 Synchronous belt drive — preliminary sizing

HTD 8M (8 mm) · z_1 = 28, z_2 = 56 · C = 450.6 mm

2.1 Design data

Profile	HTD 8M (8 mm)
Pitch p	8.00 mm
Ratio τ = z_2/z_1	2.000
Desired centre distance	C_0 = 450.0 mm
Power	P = 7.50 kW
Driver pulley speed	n_1 = 1,450 rpm
Service factor c_0	1.50

2.2 Pulleys and belt

Pulley	z	dp = z·p/π [mm]
driver (1)	28	71.30
driven (2)	56	142.60

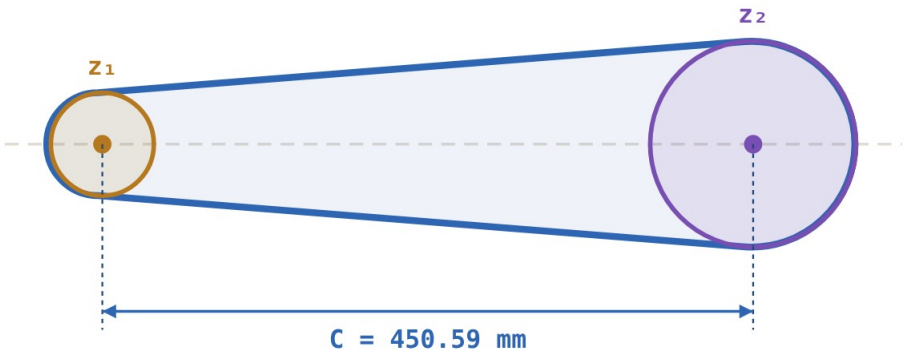
$$L_0 = 2\,C_0 + \frac{\pi}{2}\,(d_{p1} + d_{p2}) + \frac{(d_{p2} - d_{p1})^2}{4\,C_0} = 2 \cdot 450.0 + \frac{\pi}{2}\,(71.30 + 142.60) + \frac{(142.60 - 71.30)^2}{4 \cdot 450.0} = 1\,238.8$$

$$z_b = \frac{L_0}{p} = 154.85 \;\rightarrow\; 155 \;\Rightarrow\; L_p = z_b\,p = 1\,240.0 \text{ mm}$$

2.3 Actual centre distance

$$C = \frac{B + \sqrt{B^2 - 32\,(d_{p2} - d_{p1})^2}}{16}, \qquad B = 4\,L_p - 2\pi\,(d_{p1} + d_{p2}) = 3\,616.00 \text{ mm}$$

$$C = 450.59 \text{ mm} \qquad (\Delta = 0.59 \text{ mm})$$



2.4 Teeth in mesh and wrap angle

β_1	$\beta_1 = \pi - 2\arcsin\dfrac{d_{p2}-d_{p1}}{2\,C} = 170.9^\circ$
Teeth in mesh on the driver	$z_1\,\dfrac{\beta_1}{360^\circ} = 13.3$ (≥ 6: no reduction, c_1 = 1.00)

2.5 Belt width

$$v = \frac{\pi\, d_{p1}\, n_1}{60\,000} = \frac{\pi \cdot 71.30 \cdot 1\,450}{60\,000} = 5.41\ \text{m/s}$$

$$P_c = c_0\, P = 1.50 \cdot 7.50 = 11.25\ \text{kW} \ \Rightarrow\ F_t = \frac{P_c}{v} = 2\,078\ \text{N}$$

$$F_{amm} = f_0\, c_1\, c_2 = 48.0 \cdot 1.00 \cdot 1.10\ \text{N/mm} \ \Rightarrow\ b_{min} = \frac{F_t}{F_{amm}} = 39.4\ \text{mm}$$

Chosen width (catalog): **50 mm** **VERIFIED**

2.6 Dimensions for CAD and ordering

Pulley driver (1)	z_1 = 28 · dp_1 = 71.30 mm
Pulley driven (2)	z_2 = 56 · dp_2 = 142.60 mm
C	450.59 mm
Belt	Lp = 1,240.0 mm · 155 teeth · b = 50 mm
Ordering designation	HTD 8M · 1,240 mm · 155 teeth · 50 mm
Ø	Outside Ø and flanges: from the manufacturer's catalog

2.7 Assumptions and notes

- Pitch diameter $d_p = z\,p/\pi$; pitch length $L_0 = 2C_0 + \frac{\pi}{2}(d_{p1}+d_{p2}) + \frac{(d_{p2}-d_{p1})^2}{4C_0}$, rounded to a whole number of teeth.
- Actual centre distance from the exact solution of the length equation (taut belt, straight spans).
- Tangential force $F_t = P_c/v$ with $P_c = c_0\,P$; allowable force per mm of width averaged per profile, corrected with c_1 (teeth in mesh) and c_2 (length).
- The width is indicative: catalogs give the transmissible power as a function of speed and number of teeth.

Preliminary sizing: exact geometry, width from an average allowable force per profile (indicative). For the final choice use the belt manufacturer's catalog (power ratings, speed, tensioning).

Open this calculation ↗

3 Pair of cylindrical gears — preliminary sizing

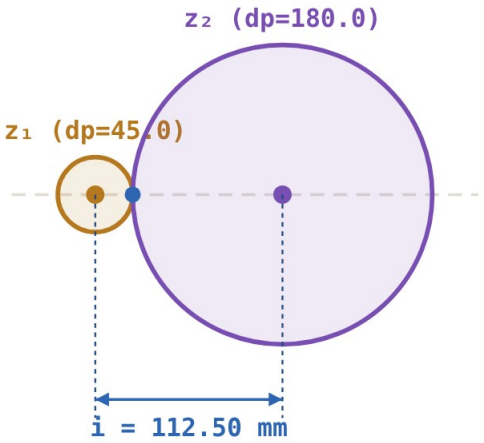
$z_1 = 20$, $z_2 = 80$, m = 2.25 mm, L = 32.8 mm

3.1 Design data

Tooth type	spur
Power	P = 7.50 kW
Pinion speed	n_1 = 725 rpm (ω_1 = 75.92 rad/s)
Pinion torque	M_1 = 98.8 N·m
Required ratio $\tau = z_1/z_2$	0.2500
Elastic coefficient of the pair Ke	35.0 GPa
Allowable contact pressure σH	900.0 MPa
Pinion profile shift x_1	0.00
Options	none

3.2 Pair geometry

Number of teeth	$z_1 = 20 \cdot z_2 = 80 \cdot \tau = 0.2500$
Module	m = 2.250 mm (UNI series 2)
Pitch diameters	dp_1 = 45.00 mm · dp_2 = 180.00 mm
Centre distance	112.50 mm
Face width L	L = 32.8 mm · ϕ = 0.729 (in the recommended range 0.5–1.0)
Undercut	z_1 = 20.0 ≥ z_min = 17.1 **VERIFIED**



3.3 Pitting design (Hertz)

$$m_{min}^3 = \frac{8\,K_e\,W\,(1+\tau)}{\omega_1 \sin 2\theta\; z_1^3\,\sigma_H^2} = \frac{8 \cdot 35\,000 \cdot 7\,500\,000 \cdot 1.2500}{75.92 \cdot 0.6428 \cdot 20^3 \cdot 900.0^2} \;\Rightarrow\; m_{min} = 2.025 \text{ mm} \quad (\phi = 1)$$

Chosen standard module: **m = 2.250 mm** (UNI series 2)

$$\phi = \frac{8\,K_e\,W\,(1+\tau)}{\omega_1 \sin 2\theta\; z_1^3\, m^3\, \sigma_H^2} = 0.729 \;\Rightarrow\; L = \phi\, d_{p1} = 0.729 \cdot 45.00 = 32.8 \text{ mm}$$

3.4 Bending check (Lewis)

$$F_c = \frac{2\,M_1}{d_{p1}} = \frac{2 \cdot 98\,786}{45.00} = 4\,390 \text{ N}$$

$$y = 0.354$$

$$\sigma_L = \frac{F_c}{L\,m\,y} = \frac{4\,390}{32.8 \cdot 2.250 \cdot 0.354} = 168.1 \text{ MPa}$$

σL = **168 MPa** ≤ 800 MPa **VERIFIED** · ϕ = **0.73** (in the recommended range 0.5–1.0)

3.5 Dimensions for CAD

Item	z	Pitch dp [mm]	Tip Ø [mm]	Root Ø [mm]
Pinion (1)	20	45.00	49.50	39.38
Wheel (2)	80	180.00	184.50	174.38

Module	m = 2.250 mm
Centre distance	112.50 mm
Face width L	32.8 mm → 33 mm
x_1 / x_2	0.00 / 0

3.6 Assumptions and notes

- Pressure angle $\theta = 20^\circ$, standard teeth (addendum m, dedendum $1.25\,m$); the pinion is the most loaded gear.
- Pitting: Hertz pressure at the pitch point, $W = \frac{\sigma_H^2\, L\, \omega_1 \sin 2\theta\; m^2 z_1^2}{8\,K_e\,(1+\tau)}$; the design fixes $\phi = L/d_{p1}$ and finds the module, then the face width at the standard module.
- Bending: Lewis formula $\sigma_L = \frac{F_c}{L\,m\,y}$, with y from the course chart (z, or the equivalent z for helical gears).
- Helical gears: factors Φ, Ψ, Γ_t from the course chart; $z_{eq} = z/\cos^3\alpha$.
- The results are preliminary: for the final design use ISO 6336 (load, speed and lubrication factors, materials).

Preliminary sizing with the method of the Machine Design course (Hertz and Lewis, factors from charts). The values are first dimensions for CAD: the final design must be checked with ISO 6336, the gear manufacturer's data or an FEM analysis.

Open this calculation ↗

4 Transmission shaft — preliminary sizing

d = 35 mm

4.1 Design data

Power	7.50 kW
Speed	725 rpm (ω = 75.92 rad/s)
Torque	Mt = 98.8 N·m (Mt = P/ω)
Material	σR = 900 MPa · σs = 700 MPa · σLF = 450 MPa
Surface finish	Medium grinding
Required safety factor	X = 2.00
Required life	infinite (≥ 10^6 cycles)
Pressure angle	θ = 20.0°
Shaft length	L = 240.0 mm
Bearings	ball (p = 3) · L = 1,000.0·10^6
Rotation seen from B	counter-clockwise

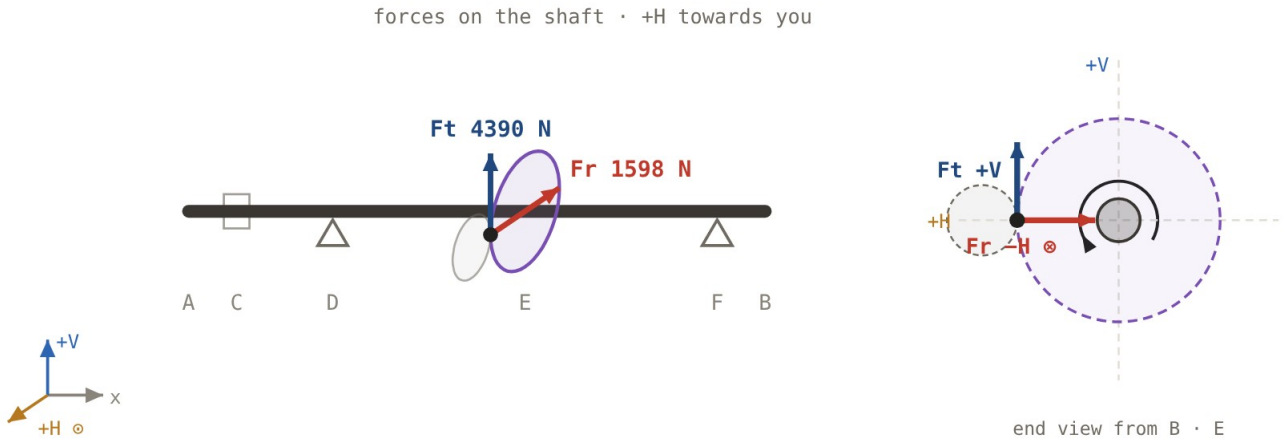
4.2 Shaft layout and loads

Point	x [mm]	Description
A	0.0	end A
C	20.0	coupling / torque, torque in · force: V 0 N, H 1,660 N, axial 0 N
D	60.0	bearing 1
E	140.0	gear pitch Ø 45.0 mm, torque out, mating gear at 90°
F	220.0	bearing 2
B	240.0	end B

Gear forces on the shaft

$$F_t = \frac{2\,M_t}{d}, \qquad F_r = \frac{F_t \tan\theta}{\cos\alpha}, \qquad F_a = F_t \tan\alpha$$

Point	Mt [N·m]	Ft [N]	Fr [N]	Fa [N]	V [N]	H [N]
E	98.8	4,390	1,598	0	4,390	-1,598



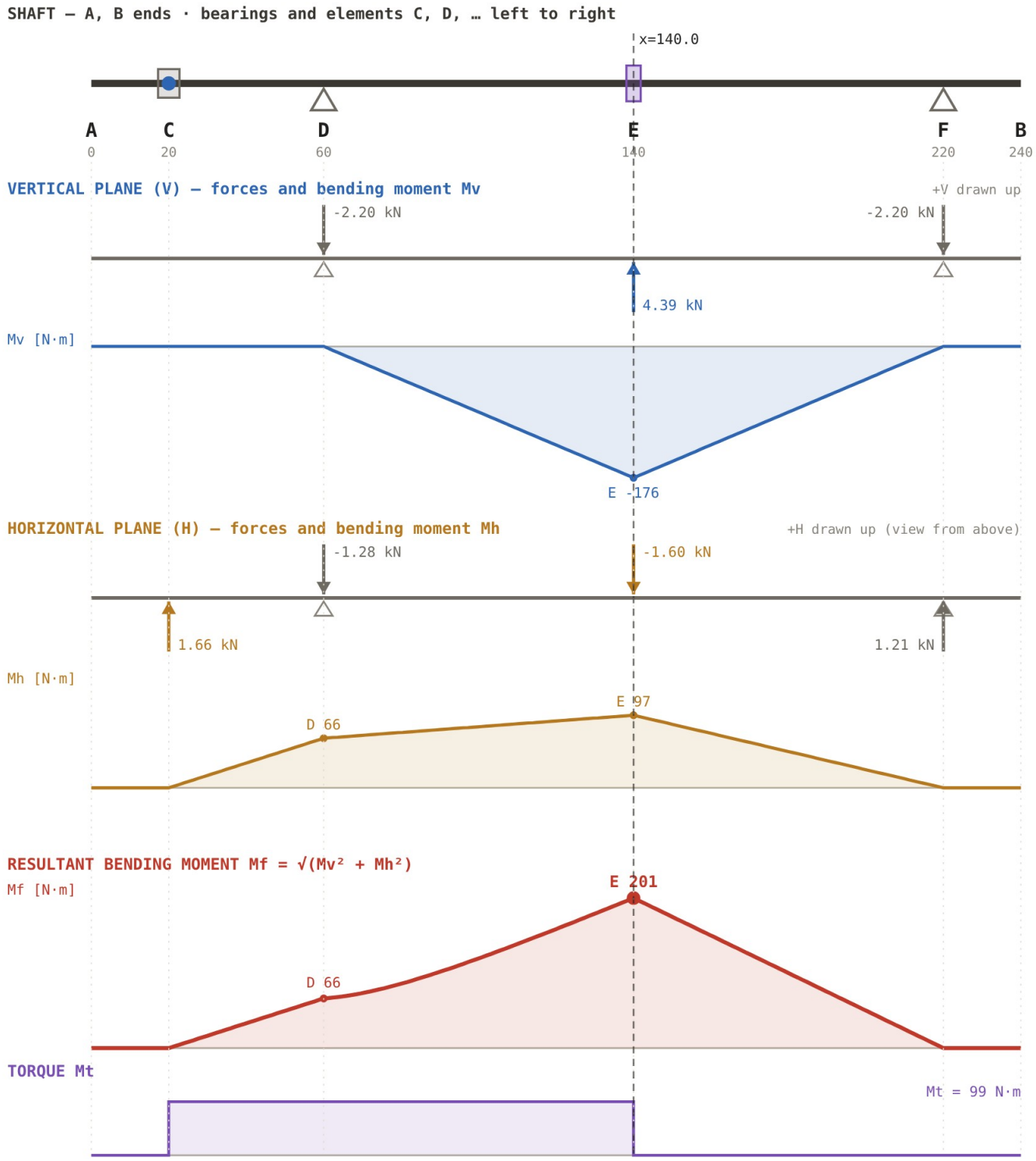
4.3 Support reactions and bearings

Point	V [kN]	H [kN]	R [kN]	axial [kN]	Required C [kN]	life with catalog C
D	-2.20	-1.28	**2.54**	—	25.39	—
F	-2.20	1.21	**2.51**	—	25.09	—

$$C = R\,L^{1/p} \qquad \left(p = 3 \text{ ball},\ p = \tfrac{10}{3} \text{ roller}\right), \qquad L = \left(\frac{C}{R}\right)^{p}$$

$$L_h = \frac{L \cdot 10^6}{60\,n} = \frac{1\,000.0 \cdot 10^6}{60 \cdot 725} = 22\,989 \text{ h}$$

4.4 Internal action diagrams



4.5 Critical section

Section with the largest Mf	x = 140.0 mm (E) · Mf = 200.7 N·m · Mt = 98.8 N·m
x	140.0 mm → Mv = -175.6, Mh = 97.1, Mf = 200.7 N·m, Mt = 98.8 N·m, N = 0 N

4.6 Section design

Designed section (data of the Design/Check tab)	Mf = 200.7 N·m · Mt = 98.8 N·m · N = 0 N · from the beam, section x = 140.0 mm
Bending / Torsion	rotating (fully reversed) / constant
Notch	shoulder + keyway · D/d = 1.20, r = 1.5 mm

Nominal stresses

$$\sigma_f = \frac{32\,M_f}{\pi\,d^3} = \frac{32 \cdot 200\,700}{\pi \cdot 35^3} = 47.68\ \text{MPa}$$

$$\tau = \frac{16\,M_t}{\pi\,d^3} = \frac{16 \cdot 98\,786}{\pi \cdot 35^3} = 11.73\ \text{MPa} \qquad (\tau_a = 0.00,\ \tau_m = 11.73)$$

Coefficients

$$K_e = q\,(K_t - 1) + 1 = 0.833 \cdot (2.056 - 1) + 1 = 1.880 \qquad \left(\tfrac{r}{d} = 0.0429,\ \tfrac{D}{d} = 1.20\right)$$

$$K_e = K_{e,\text{sp}} \cdot k_{e,\text{l}} = 1.880 \cdot 1.60 = 3.008, \qquad K'_e = 1.560 \cdot 1.30 = 2.028$$

$$b_1(35) = 0.811, \qquad b_2 = 0.875$$

$$\sigma_N = \sigma_{LF} = 450\ \text{MPa}$$

Equivalent stresses

$$\sigma_{a,eq} = \sqrt{(K_e\,\sigma_a)^2 + 3\,(K'_e\,\tau_a)^2} = \sqrt{(3.008 \cdot 47.68)^2 + 3\,(2.028 \cdot 0.00)^2} = 143.44\ \text{MPa}$$

$$\sigma_{m,eq} = \frac{\sigma_m}{2} + \sqrt{\left(\frac{\sigma_m}{2}\right)^2 + \tau_m^2} = \frac{0.00}{2} + \sqrt{\left(\frac{0.00}{2}\right)^2 + 11.73^2} = 11.73\ \text{MPa}$$

Goodman line · Yield

$$\frac{1}{X} = \frac{\sigma_{a,eq}}{b_1\,b_2\,\sigma_N} + \frac{\sigma_{m,eq}}{\sigma_R} = \frac{143.44}{0.811 \cdot 0.875 \cdot 450.0} + \frac{11.73}{900} \ \Rightarrow\ X = 2.163$$

$$X = \frac{\sigma_s}{\sigma_{a,eq} + \sigma_{m,eq}} = \frac{700}{155.17} = 4.511$$

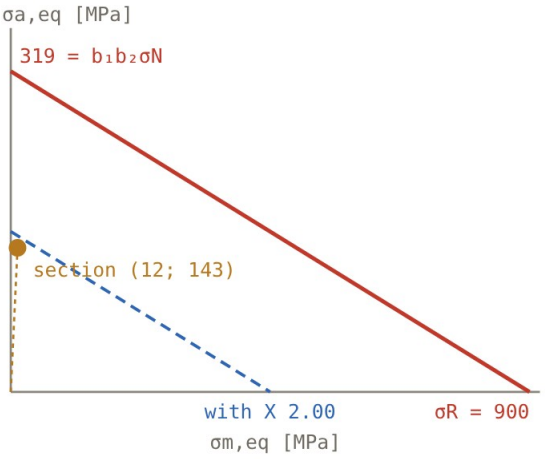
static von Mises on the peaks (no Kt):

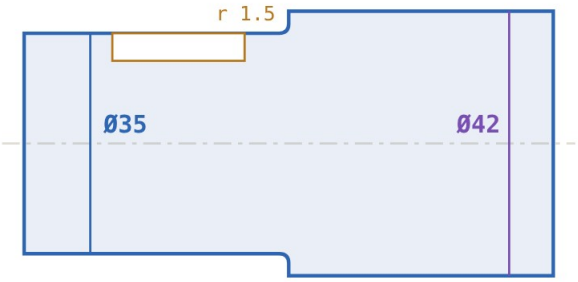
$$\sigma_{id} = \sqrt{\sigma^2 + 3\,\tau^2} = 51.83\ \text{MPa} \ \Rightarrow\ X = \frac{\sigma_s}{\sigma_{id}} = 13.51$$

Minimum diameter: **d ≥ 34.0 mm** (governed by fatigue)

Chosen diameter (bearing bore): **d = 35 mm · D = 42 mm** · r = 1.5 mm

X fatigue = **2.16 VERIFIED** · X yield = **4.51 VERIFIED** (X ≥ 2.00)





4.7 Dimensions for CAD

Item	Value	Note
Shaft length	L = 240.0 mm	
bearing 1 (D)	x = 60.0 mm	
bearing 2 (F)	x = 220.0 mm	
coupling / torque (C)	x = 20.0 mm	
force (C)	x = 20.0 mm	
gear (E)	x = 140.0 mm · pitch Ø 45.0 mm	
Designed section (data of the Design/Check tab)	d = 35 mm · D = 42 mm · r = 1.5 mm	shoulder + keyway

4.8 Assumptions and notes

- Shaft on two supports (bearings) in two orthogonal planes V and H; resultant bending moment $M_f = \sqrt{M_v^2 + M_h^2}$.
- Gear forces applied at the mesh point; the axial force of helical gears adds a concentrated couple $F_a\, r$.
- Fatigue: $\sigma_{a,eq} = \sqrt{(K_e\,\sigma_a)^2 + 3\,(K'_e\,\tau_a)^2}$, $\sigma_{m,eq} = \frac{\sigma_m}{2} + \sqrt{\left(\frac{\sigma_m}{2}\right)^2 + \tau_m^2}$; Goodman line $\frac{\sigma_{a,eq}}{b_1 b_2 \sigma_N} + \frac{\sigma_{m,eq}}{\sigma_R} = \frac{1}{X}$.
- K_t from the course formula $B\,(r/d)^a$, q from Neuber's formula (it matches the charts), $K_e = q\,(K_t - 1) + 1$; b_1 and b_2 from the course charts.
- Finite life: Wöhler line between σ_R at 10^3 and σ_{LF} at 10^6 cycles.

Preliminary sizing with the method of the Machine Design course (Goodman line, coefficients from charts). The values are first dimensions for CAD: the final check must follow the standards (e.g. DIN 743 / ISO 6336), the manufacturers' catalogs or an FEM analysis.

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5 Shaft-hole fit ISO 286

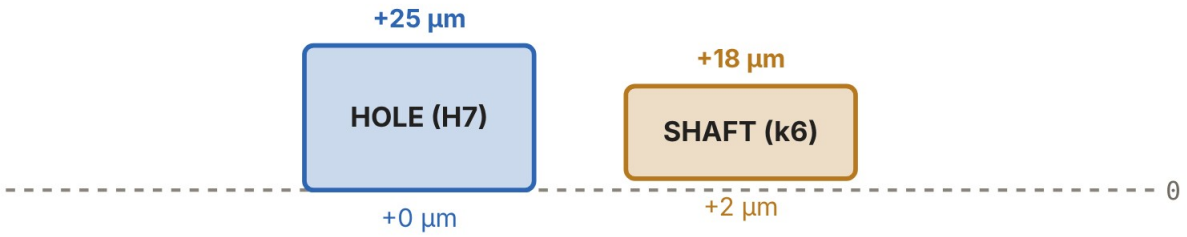
Ø35.00 H7/k6 · transition

5.1 Data

Nominal diameter	Ø 35.00 mm
ISO 286 size range	30 – 50 mm
Fit	H7/k6 — Light press / Hand press fit (H7/k6)

5.2 Deviations and limit sizes

	Upper dev.	Lower dev.	Tolerance	Max size	Min size
Hole H7	+25 µm	0 µm	25 µm	35.025	35.000
Shaft k6	+18 µm	+2 µm	16 µm	35.018	35.002



5.3 Clearances and type of fit

$$G_{max} = ES - ei = 25.0 - (2.0) = 23.0\ \mu\text{m}$$
$$G_{min} = EI - es = 0.0 - (18.0) = -18.0\ \mu\text{m}$$
$$G_{med} = \frac{G_{max} + G_{min}}{2} = 2.5\ \mu\text{m}$$

Type: **transition** · Min clearance -18 µm · Max clearance +23 µm

5.4 Machining and roughness

	Machining	Roughness
Hole	Finish Reaming	Ra 1.6 µm / 63 µin
Shaft	Cylindrical Grinding	Ra 0.8 µm / 32 µin

5.5 Comparison with the preferred fits of the same system

Fit	Min clearance	Max clearance	Type
H11/c11	+120 µm	+440 µm	clearance
H9/d9	+80 µm	+204 µm	clearance
H8/f7	+25 µm	+89 µm	clearance
H7/f7	+25 µm	+75 µm	clearance
H7/g6	+9 µm	+50 µm	clearance
H7/h6	0 µm	+41 µm	clearance
H8/h7	0 µm	+64 µm	clearance
H7/js6	-8 µm	+33 µm	transition
H7/k6	**-18 µm**	**+23 µm**	**transition**
H7/n6	-33 µm	+8 µm	transition
H7/p6	-42 µm	-1 µm	interference
H7/r6	-50 µm	-9 µm	interference
H7/s6	-59 µm	-18 µm	interference

5.6 Drawing callouts

Hole	Ø35 H7 (+0.025 / +0.000)
Shaft	Ø35 k6 (+0.018 / +0.002)
Drawing callout	Ø35 H7/k6

5.7 Notes

- Hole-basis system: hole H (EI = 0), the shaft carries the tolerance position.
- Clearance = hole size − shaft size; a negative clearance is an interference.
- Machining and roughness are typical indications for the tolerance grade.

Values from ISO 286-2 for the size range of the nominal diameter (up to 500 mm). For interference fits also check the fit pressure and stresses (Lamé) and the assembly conditions.

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Preliminary sizing with the method of the Machine Design course (Goodman line, coefficients from charts). The values are first dimensions for CAD: the final check must follow the standards (e.g. DIN 743 / ISO 6336), the manufacturers' catalogs or an FEM analysis.

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